

AKSHAY S.

Chief of Staff / Technical Strategy — Physical AI, Robotics & Distributed Energy Systems

Technical architecture and executive strategy operating as one function — pressure-testing engineering roadmaps, modeling deployment unit economics, and turning core physics into capital-allocation decisions.

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INDEPENDENT SYSTEMS RESEARCH

Physical AI, Energy & Distributed Architecture (2025–Present)

The Architecture of Abundance: How Energy Flows and Creates Intelligence

- **Modeled terrestrial RF power beaming** — link budgets against the FCC's 10 W/m² exposure ceiling, inverse-square path loss, and space-based microwave transmission geometry.
- **Derived the 25.8-hour mass recursion wall** — the point at which added battery mass in humanoid robots stops extending runtime because it demands more structure and torque to carry itself.
- **Proposed a trickle-spike power architecture** — continuous ambient RF harvest paired with supercapacitors for transient torque spikes — cutting baseline locomotion/idle draw by ~30% in the model.

Computing Leaves the Device: Ambient Power & the Spatial Internet

- **Modeled sub-100 mW battery-free spatial interfaces** — offloading rendering to distributed neighborhood micro-nodes within a strict 20 ms motion-to-photon latency budget.
- **Designed a frame-by-frame decision rule** — choosing, in real time, whether to send the raw visual data or just regenerate the scene locally — whichever costs less power without breaking the latency budget.

Ambient-Powered Computing: The Four-Curve Convergence

- **Systems-level synthesis** — tracing the intersection of falling semiconductor power draw, rising rectenna efficiency, hyperscaler compute capex, and reusable orbital launch economics to time the arrival of ambient-powered compute.

TECHNICAL STRATEGY & OPERATIONAL DILIGENCE

Hardware Systems & Infrastructure Analysis | 2025–2026

- **Deconstructed hardware trade-offs** — regulatory exposure, and deployment unit economics for wide-area wireless power and localized compute nodes.
- **Pitched strategic architecture briefs directly to venture capital funds** — translating core engineering constraints into investable theses.

EDUCATION, TECHNICAL COMPETENCIES & LEADERSHIP

Indian Institute of Technology (IIT) Madras — B.Tech, Chemical Engineering (2020–2025)

Technical Competencies: Systems modeling from physical fundamentals · thermodynamic & mass/energy transport analysis · RF/power link budgets · hardware thermal & power modeling · numerical modeling (Python) · technical diligence.

Athletics & Leadership: Vice-Captain, IIT Madras Table Tennis Team — 2× Inter-IIT Gold Medalist (2022, 2023). Mentor, Saathi Wellness Program.